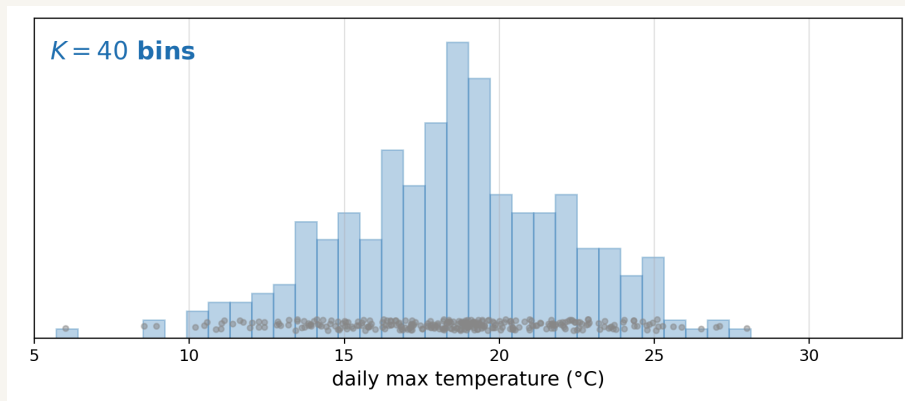


Multivariate Gaussian models

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Summarizing data

A year of temperatures in Rabat



300 numbers. How do you *summarize* them, with as few numbers as possible ?

Level 1: the mean (or median)

Level 2: add variability: the standard deviation

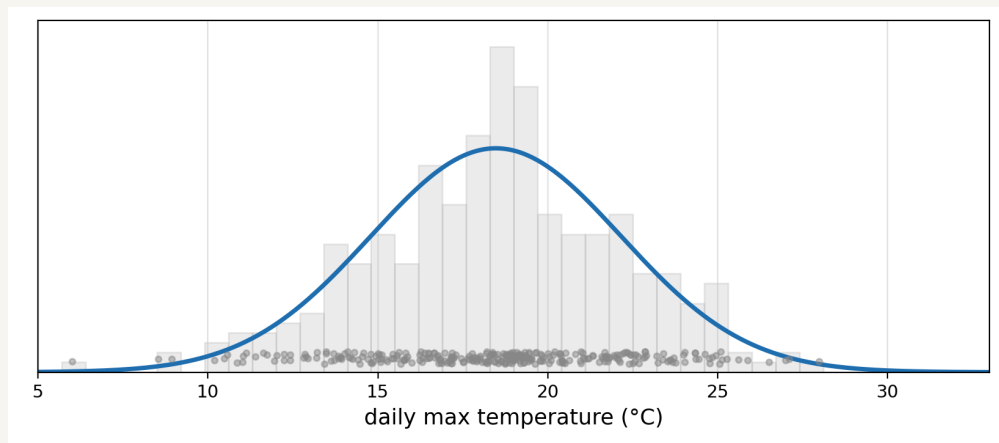
Level 3: a histogram, the whole shape

What information did we gain with level 3?

With how many *summary* numbers ?

One height per bin: K bins, K numbers

A year of temperatures in Rabat



Level 1: the mean

Level 2: the standard deviation

Level 3: a histogram

Level 4: a parametric distribution

Can we *approximate* this shape with fewer numbers ?

Using a **Gaussian** model $\mathcal{N}(\bar{x}, s^2)$

Histograms are considered **non-parametric** estimators.

Now two cities

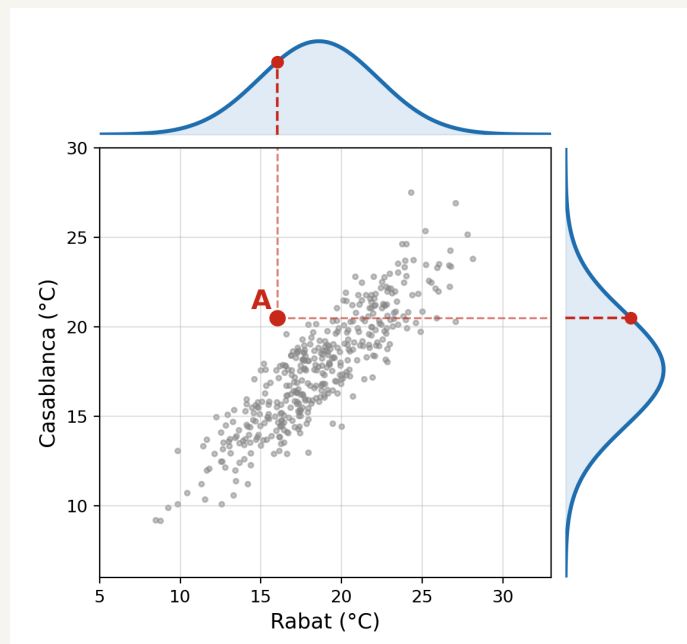
day	Rabat (°C)	Casablanca (°C)
1	16.0	15.8
2	20.4	19.3
⋮	⋮	⋮

One Gaussian per city: $X_j \sim \mathcal{N}(\bar{x}_j, s_j^2)$

What would this model be bad at ?

A = (16, 20.5): likely for each city alone, unlikely jointly.

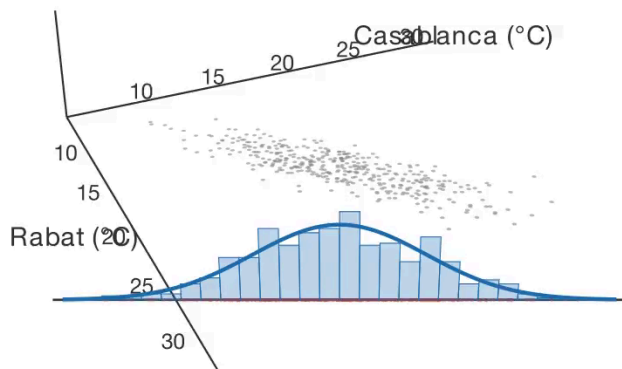
It misses the **correlation**.



We need a **Gaussian in dimension** $d > 1$: center μ , covariance $\Sigma = \begin{pmatrix} \text{var}_1 & \text{cov} \\ \text{cov} & \text{var}_2 \end{pmatrix}$

The multivariate Gaussian

How do we define a Gaussian vector ?



If we projected on each axis we get a **Gaussian** in 1D

“ $\mathbf{X} = (X_1, X_2)$ is a Gaussian vector if and only if X_1 and X_2 are univariate Gaussians”

Is that enough ?

No: a distribution should not depend on the basis we choose.

What we want: a Gaussian seen from **every** direction.

A Gaussian random vector

Definition: $\mathbf{X} = (X_1, \dots, X_d)^\top$ is a **Gaussian vector** when every linear combination $\mathbf{a}^\top \mathbf{X} = a_1 X_1 + \dots + a_d X_d$ follows a one-dimensional Gaussian.

Exercise Let $\mathbf{X}$ be a Gaussian vector with mean $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$, covariance $\boldsymbol{\Sigma}$, and $\mathbf{a} \in \mathbb{R}^d$: what are $\mathbb{E}[\mathbf{a}^\top \mathbf{X}]$, $\mathbb{V}(\mathbf{a}^\top \mathbf{X})$? Determine the distribution of $\mathbf{a}^\top \mathbf{X}$.

The standard Gaussian

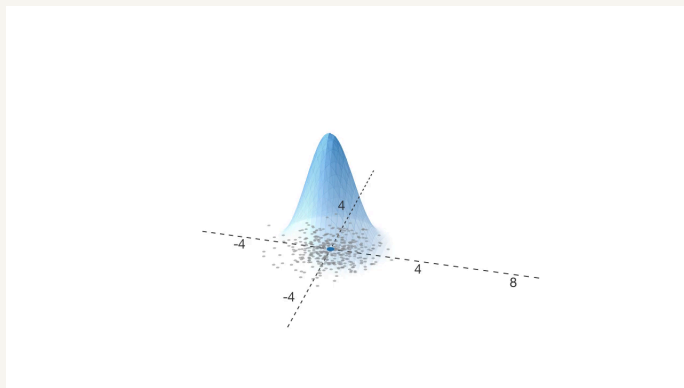
Start simple: $Z_1, \dots, Z_d$ **independent** $\mathcal{N}(0, 1)$

The joint vector $\mathbf{Z} = (Z_1, \dots, Z_d)^\top$

Is $\mathbf{Z}$ a Gaussian vector ? What is its density ?

$$f_{\mathbf{Z}}(\mathbf{z}) = \prod_{i=1}^d \frac{1}{\sqrt{2\pi}} e^{-z_i^2/2} = \frac{1}{(2\pi)^{d/2}} e^{-\frac{1}{2}\|\mathbf{z}\|^2}$$

Yes: any $\mathbf{a}^\top \mathbf{Z}$ is a sum of independent Gaussians. $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_d)$, the **standard** Gaussian.



Here $d = 2$

Transform it

$$\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_d)$$

Under what simple transformations does $\mathbf{Z}$ remain a Gaussian vector ?

$$\mathbf{X} = \mathbf{A}\mathbf{Z} + \boldsymbol{\mu}, \quad \mathbf{A} \in \mathbb{R}^{d \times d}, \quad \boldsymbol{\mu} \in \mathbb{R}^d$$

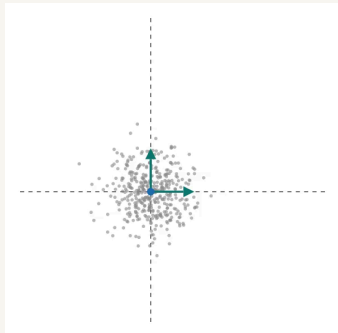
Mean and covariance of $\mathbf{X}$?

$$\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}, \quad \text{Cov}(\mathbf{X}) = \mathbf{A}\mathbf{A}^\top$$

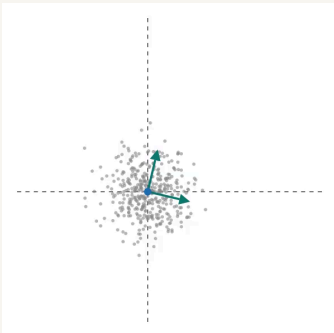
$$\text{With } \boldsymbol{\Sigma} \stackrel{\text{def}}{=} \mathbf{A}\mathbf{A}^\top: \quad \mathbf{X} = \mathbf{A}\mathbf{Z} + \boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

What does $AZ + \mu$ do ?

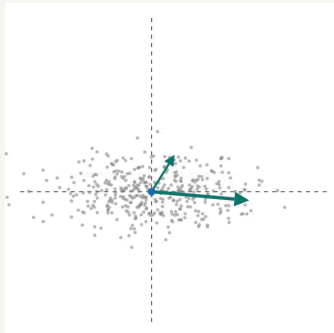
What geometric transformations does $AZ + \mu$ apply to the cloud ?



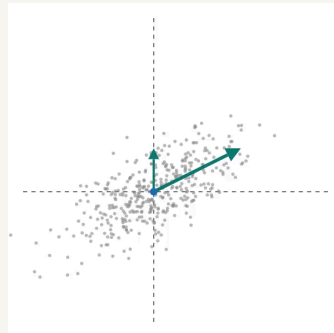
Z



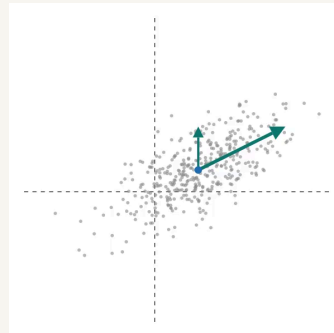
rotate: V^T



stretch: S



rotate: U



shift: $+\mu$

$A = USV^T$ (SVD): **rotate, stretch, rotate**, then shift. Here $A = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$, $\mu = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

Exercise 1

Change of variables

Let $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_d)$ and $\mathbf{X} = \mathbf{AZ} + \boldsymbol{\mu}$ with $\mathbf{A}$ invertible, and $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^\top$.

Express $\mathbf{z}$ as a function of $\mathbf{x}$, then show that $\|\mathbf{z}\|^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$.

The density

If Σ is **invertible**, a Gaussian vector of $\mathbb{R}^d$ admits a density:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})\right), \quad \mathbf{x} \in \mathbb{R}^d$$

How many numbers does it take to characterize a Gaussian of $\mathbb{R}^d$?

d numbers for $\boldsymbol{\mu}$, $\frac{d(d+1)}{2}$ for Σ

$d = 1, \Sigma = \sigma^2$: $f(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$, the familiar 1D Gaussian.

Correlation and independence?

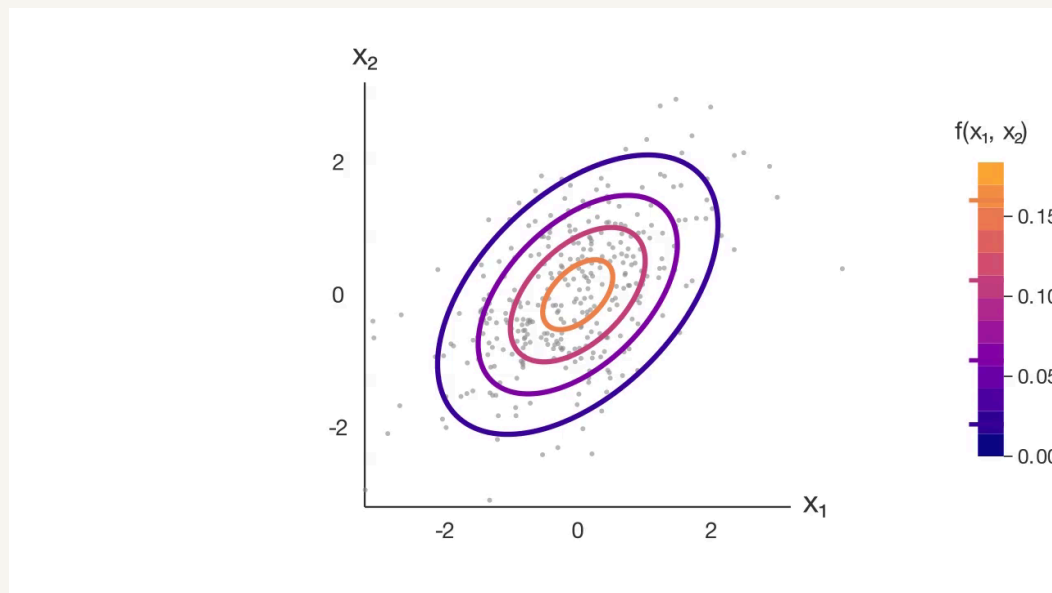
Take Σ diagonal (all correlations zero): what does the density become ?

$$\Sigma = \begin{pmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_d^2 \end{pmatrix} \implies (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = \frac{(x_1 - \mu_1)^2}{\sigma_1^2} + \dots + \frac{(x_d - \mu_d)^2}{\sigma_d^2}$$
$$f(\mathbf{x}) = \prod_{i=1}^d f_i(x_i) \quad \text{the density factorizes: } X_1, \dots, X_d \text{ independent}$$

For Gaussian vectors: **zero correlation** $\iff$ **independence**.

... but only $\Leftarrow$ in general

Reading a density



The density in 3D over the cloud, and the curves where it is **constant**

A density drawn in 3D is hard to read.

Instead, draw only the curves where it is constant: the **level sets**.

The geometry of Σ

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

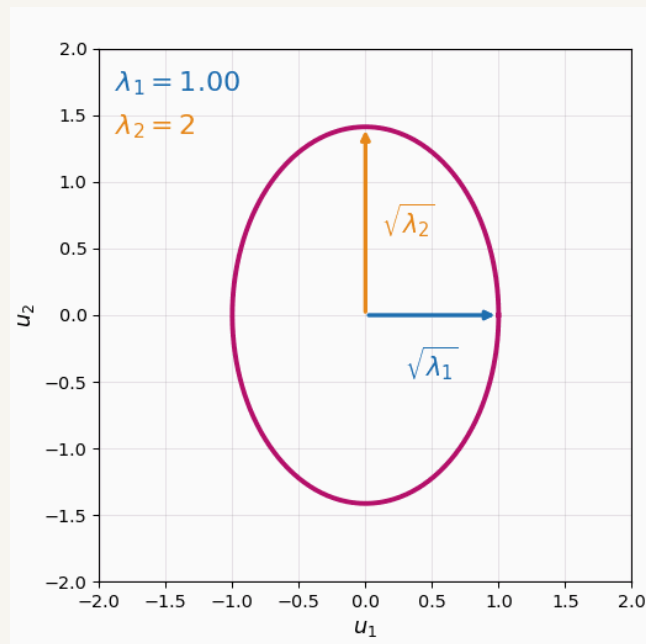
On what $\mathbf{x}$ does this density remain constant ?

Level set (*Courbe de niveau*) at height c :

$$\mathcal{E}_c = \{\mathbf{x} \in \mathbb{R}^d : (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c\}.$$

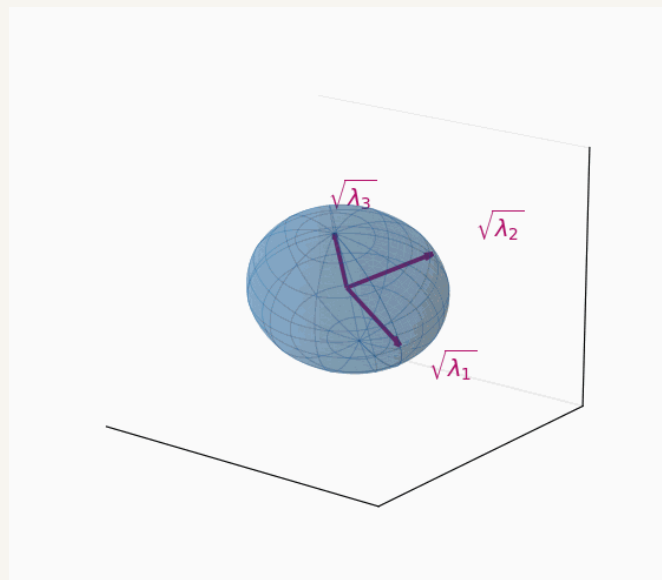
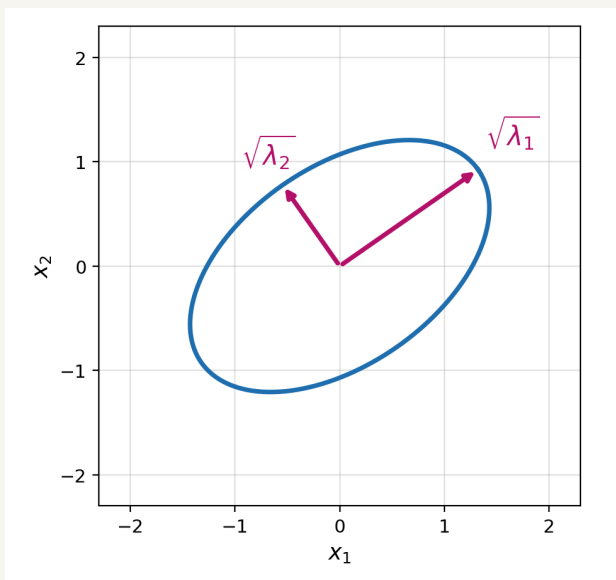
Exercise Take $d = 2$. Analyze and simplify $\mathcal{E}_c$.

(Whiteboard)



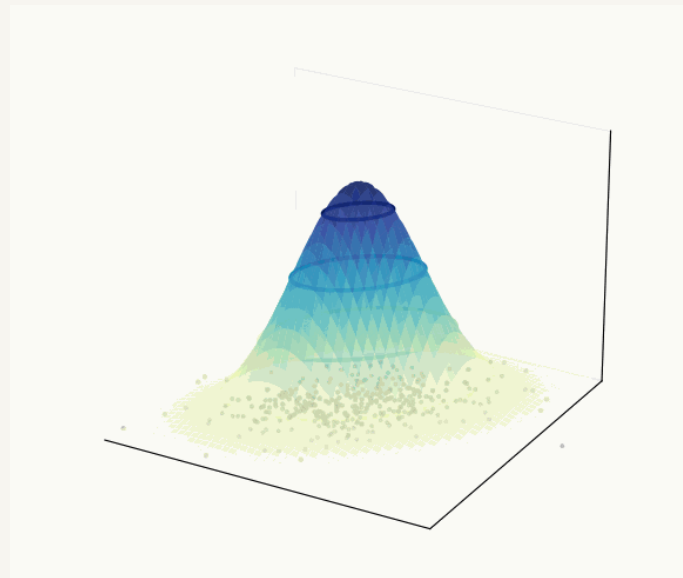
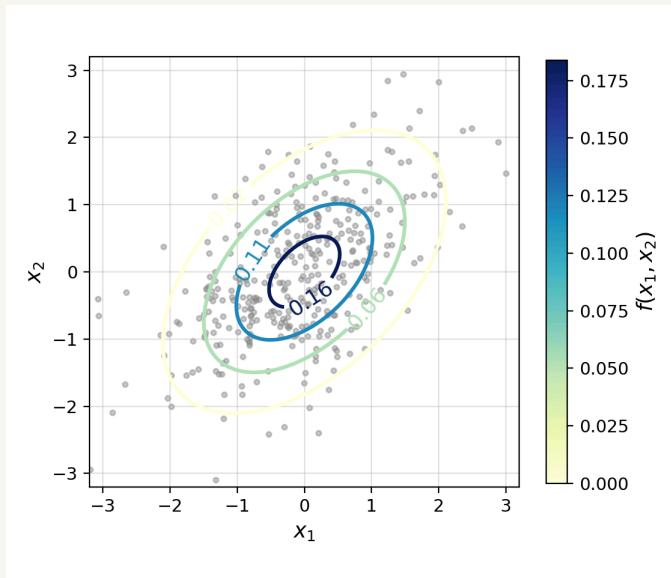
Ellipsoids in any dimension

In dimension d : the level set $\{\mathbf{x} : (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) = c\}$ is an **ellipsoid** centered at $\boldsymbol{\mu}$: its axes are the eigenvectors of $\boldsymbol{\Sigma}$, with radius $\sqrt{c \lambda_i}$ along the i -th axis.



Level sets

$$\mathbf{x} \sim \mathcal{N}\left(\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}\right)$$



Gaussian geometry

Mahalanobis distance

	Rabat (°C)	Casablanca (°C)
mean μ	19	18
A	23	22
B	23	14

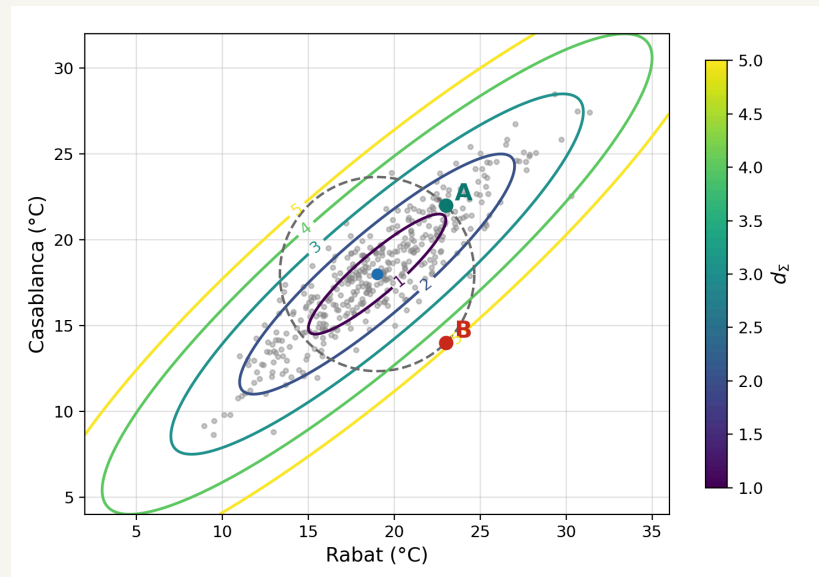
Which is *further* from the mean **A** or **B**?

$$\|A - \mu\| = \|B - \mu\| = \sqrt{32}$$

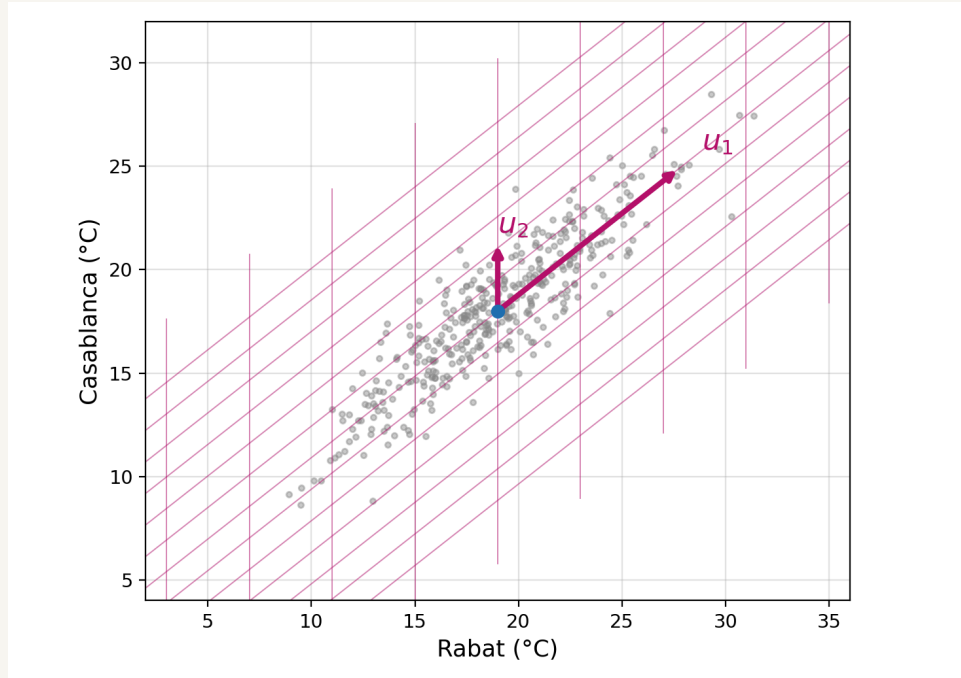
But are they *equally likely* as observations ?

Comparing the density values is equivalent to comparing their level sets: $(\mathbf{x} - \mu)^\top \Sigma^{-1}(\mathbf{x} - \mu)$

$d_\Sigma(\mu, \mathbf{x}) \stackrel{\text{def}}{=} \sqrt{(\mathbf{x} - \mu)^\top \Sigma^{-1}(\mathbf{x} - \mu)}$ is a *real distance*. Here $d_\Sigma(\mu, A) = 1.1$, ordinary; $d_\Sigma(\mu, B) = 4.8$, an **anomaly**.



A new geometry



Σ defines a new scalar product $\langle u, v \rangle_{\Sigma} = u^{\top} \Sigma^{-1} v$ (see section 1.4 in the textbook).

Fitting the model and regularization

Maximum likelihood

Same recipe as in 1D: maximize $\prod_{i=1}^n f(\mathbf{x}_i)$ over $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \quad \hat{\boldsymbol{\Sigma}} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \hat{\boldsymbol{\mu}})(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^\top$$

- $\hat{\boldsymbol{\mu}}$: d averages, easy and accurate
- $\hat{\boldsymbol{\Sigma}}$: $\frac{d(d+1)}{2}$ parameters from n points...

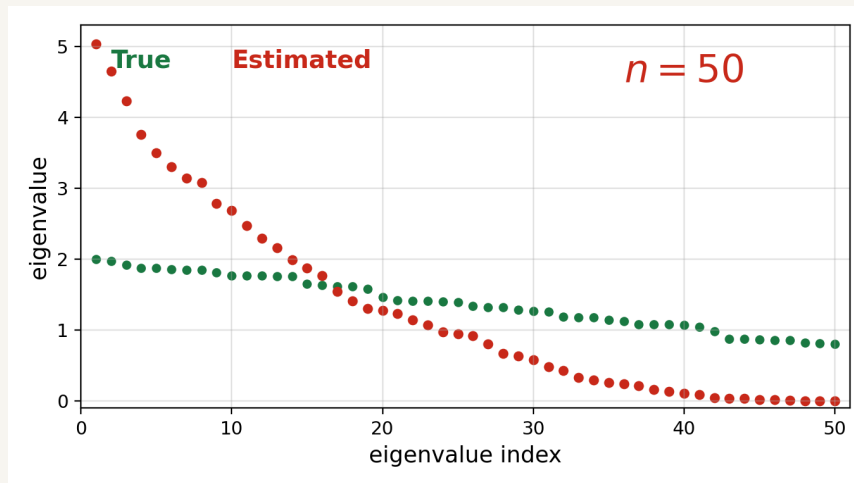
At $d = 50$, that is 1275 parameters. With $n = 100$ points: what could go wrong?

Exercise: show that $\text{rank}(\hat{\boldsymbol{\Sigma}}) \leq \min(n - 1, d)$. What does that force as soon as $n \leq d$?



This is the maximizer when $\hat{\boldsymbol{\Sigma}}$ is invertible; if not, the likelihood is unbounded and has no maximizer.

Estimating Σ is hard



True Σ : diagonal, $d = 50$ eigenvalues. How do you expect the eigenvalues of $\hat{\Sigma}$ to be ?

And if we decrease n ?

At $n = 50 = d$ the estimate $\hat{\Sigma}$ is **singular**: too little data for all $\frac{d(d-1)}{2}$ correlations.

The cure: regularization

We introduce our own *bias*:

$$\hat{\Sigma}_{\lambda} = \hat{\Sigma} + \lambda I, \quad \lambda > 0$$

Eigenvalues of $\hat{\Sigma}_{\lambda}$?

$\hat{\Sigma}_{\lambda}$ is invertible; at the expense of what ?

Every variance is inflated; unbounded in λ .

What about this new formula ?

$$\hat{\Sigma}_{\lambda} = (1 - \lambda) \hat{\Sigma} + \lambda I, \quad \lambda \in [0, 1]$$

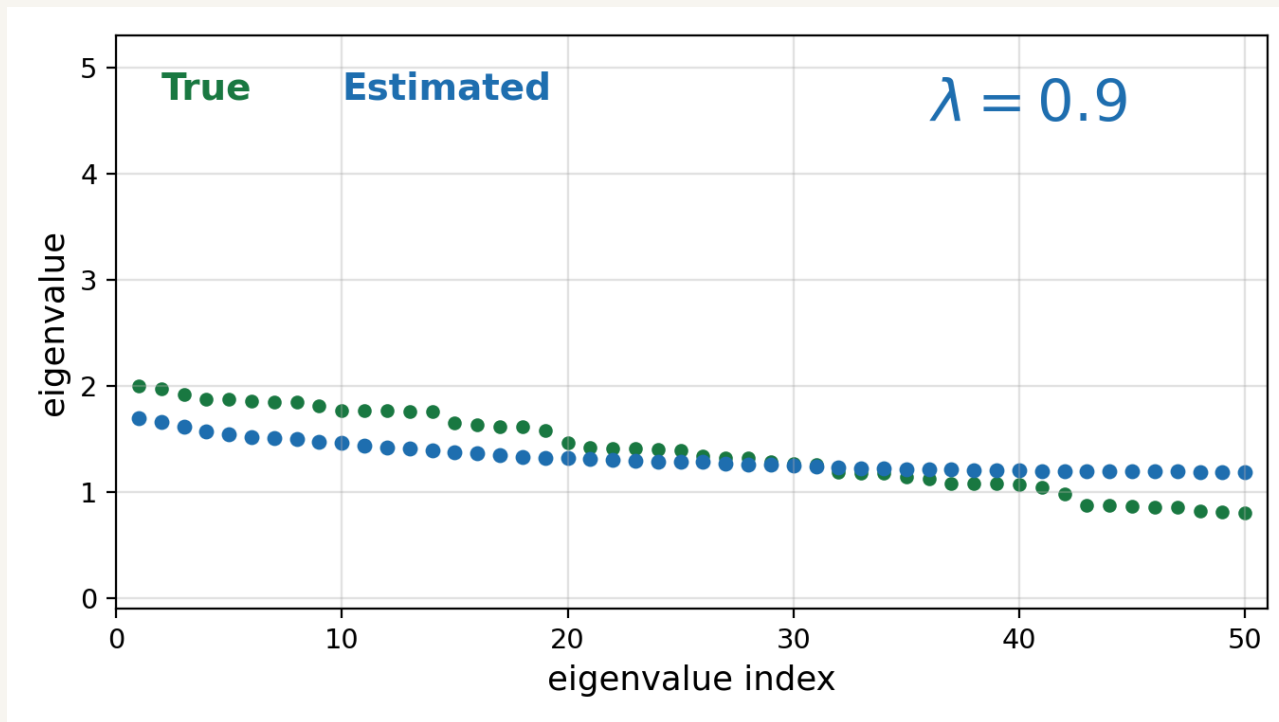
Total variance stays bounded. But what happens as $\lambda \rightarrow 1$?

$\hat{\Sigma}_{\lambda} \rightarrow I$: every variance pulled to 1, **regardless of the true variance.**

$$\hat{\Sigma}_{\lambda} = (1 - \lambda) \hat{\Sigma} + \lambda \tau I, \quad \tau = \frac{\text{tr}(\hat{\Sigma})}{d} \quad (\text{the average variance})$$

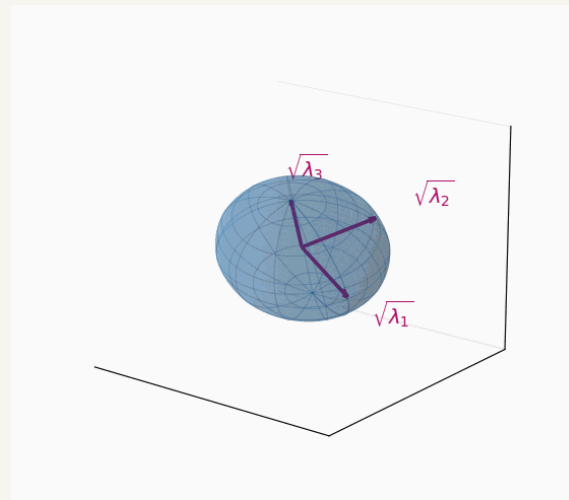
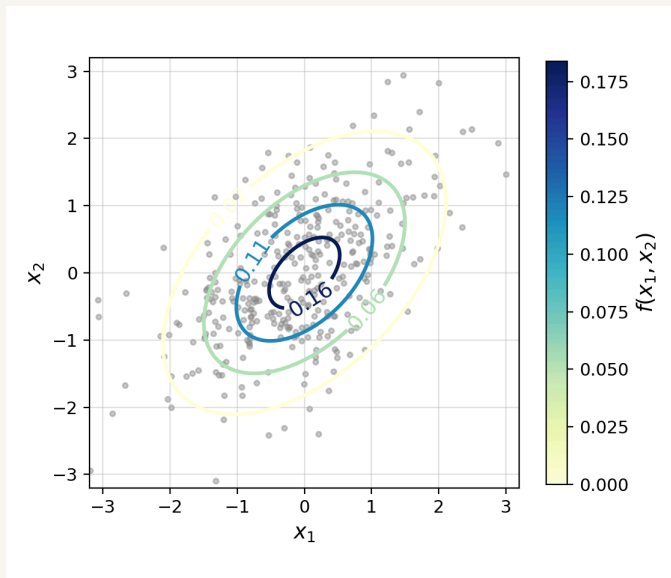
Now $\text{tr}(\hat{\Sigma}_{\lambda}) = \text{tr}(\hat{\Sigma})$ for every λ : the total variance is exactly preserved.

Shrinkage in action



λ is the **shrinkage parameter**: variances shrink towards τ , correlations towards zero.

What to remember



The Gaussian is the **simplest model of a cluster**: a center μ and an ellipsoidal shape Σ .

But in high dimension that shape has $\frac{d(d+1)}{2}$ parameters: estimating it accurately demands a **lot** of samples: the *curse of dimension*.

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